Deficiency

A prolonged deficiency of iodine causes enlargement of the thyroid gland, known as goiter.

Fluoride

Although some scientists have doubted that fluoride is an essential nutrient for humans, this trace mineral is currently considered essential. The main function of fluoride is to harden the bones and teeth.

Food sources

The only good dietary sources of fluoride are fluoridated water, seafood, seaweed, and tea. Some natural water sources are also high in fluoride.

Deficiency

Deficiency of fluoride results in the development of dental caries.

Manganese

The trace mineral manganese is required for normal brain function. This micro mineral also aids in the synthesis of collagen, urea, fatty acids, and cholesterol. Manganese is involved in bone growth and is required for the digestion of protein.

Food sources

Manganese is present in many vegetable foods, and our diets supply plenty to meet the requirement, which is minimal.

Deficiency

For the above reason, deficiency of this trace mineral has not been observed in humans, although it can be induced in experimental animals.

Selenium

The best understood role of the micro mineral selenium is its involvement in an enzyme system that helps to protect cell membranes against oxidative damage. Thus selenium has an effect similar to that of the antioxidant vitamin E.

Food sources

Selenium is widely distributed in both animals and plant foods.

Deficiency

Deficiency resulting from low dietary intake is unlikely because of its wide distribution in animals and plant foods.

Chromium

The trace mineral chromium is involved in the normal use of glucose and blood lipids and in the functioning of insulin. The active form of chromium in the body is called the Glucose Tolerance Factor (GTF).

Molybdenum

Molybdenum is required for several enzyme systems in the body. One of these enzyme systems involved in the formation of uric acid, a waste product of protein metabolism is excreted in the urine.

Food sources

The molybdenum of plant foods depends on the type of soil in which they are grown. (Neutral or alkaline soil).

Deficiency

Molybdenum deficiency has not been observed in human beings or any other species.

Ultra trace mineral

Seven minerals present in foods and in human bodies have been found to be essential nutrients for animals and may be essential for humans, but nutritional requirements for them have not yet been clearly defined. This group of minerals sometimes called ultra-trace minerals, includes arsenic, boron, cobalt, nickel, silicon, tin, and vanadium.

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